

Vipul Sharma

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Aspiring Data/AI Engineer with strong CS fundamentals in database design, SQL/NoSQL querying, backend API development, and applied machine learning/GenAI. Comfortable end-to-end, from data modeling to containerized deployment. GATE CS 2025 qualified (Top 5.5%); published researcher.

Education

Degree	Institute/Board	CGPA/%	Year
M.Tech – CSE	Indian Institute of Information Technology, Una	8.65 (Current)	2025–2027
B.Tech – CSE	Himachal Pradesh Technical University	8.16	2021–2025
Senior Secondary	CBSE Board	94.8%	2021
Secondary	HP Board	92.4%	2019

Technical Skills

Languages & Databases: Python, SQL, C++, JavaScript • MySQL, PostgreSQL, MongoDB (schema design, querying, aggregation pipelines)

Data & Backend: REST API Design, Data Modeling, Data Validation & Processing, Graph Modeling, Node.js, Express.js

Machine Learning & AI: NumPy, Pandas, Scikit-learn, Applied Machine Learning, Algorithm Design, Data Analysis, Generative AI Fundamentals, Prompt Engineering, LLM Basics

Tools & Foundations: Docker, Git, Linux, VS Code • DBMS, DSA, Operating Systems, Computer Architecture, Compiler Design, Computer Networks

Projects

North Stars – Travel Data Platform for Dhauladhar 2025

B.Tech Final Year Project (Team)

- Designed and normalized MongoDB schemas for places, routes, and user data to support fast, consistent querying across a location-based dataset.
- Built aggregation-pipeline-driven APIs to filter, search, and serve location/POI data efficiently, and containerized the app with Docker.

QuantX AI – Cybersecurity Intelligence Platform 2026

AI Platform

- Built an AI-based platform that ingests and analyzes threat-intelligence data, designing data-ingestion flows to structure raw data into a queryable, analysis-ready form.
- Developed a real-time phishing-detection module that parses URL and page-content data on the fly, with lightweight feature extraction for low-latency classification.

VIPVISUALIZE – Kidney-Exchange Compatibility & Matching 2025

Python

- Modeled kidney-exchange data as a compatibility graph (donor–patient pairs as nodes, compatibility as edges) and implemented/benchmarked matching algorithms, including a custom approach, to find optimal exchange cycles and chains.
- Built an interactive visualization and deployed it on Hugging Face for exploring graphs and algorithm outputs.

Experience

Teaching Assistant – IIIT Una 2026 – Present

- Conduct hands-on lab practicals and evaluate coursework for Network Security and Object-Oriented Programming (C++), supporting undergraduate students with implementation and debugging.

Excellence Technology – Trainee Jan 2025 – May 2025

- Contributed to development and support tasks as part of a hands-on technical training program, gaining practical industry exposure.

Freelance & Technical Contributions – Remote 2023 – Present

- Prepared technical reports and documentation, and debugged/reviewed C/C++, Python, and web/database-driven projects to improve code correctness and reliability.

Achievements

- GATE CS 2025 Qualified** – Top 5.5% among 1,70,000+ candidates.
- Research Publication** – *Cyber Attacks and Prevention through Past Learnings*, view paper (2024).

Certifications & Relevant Coursework

Certifications: IBM Cyber Security Fundamentals, Generative AI (LinkedIn Learning), MERN Stack Development, Web Dev (Internshala), Android Dev (Internshala), NPTEL ×2 (one with Elite/Topper distinction)

Coursework: DBMS, Data Structures & Algorithms, Operating Systems, Computer Networks, Compiler Design, Probability & Statistics, Linear Algebra

Positions of Responsibility

Head, Event Management Team · **Position Holder**, Technical & Cultural Clubs

2021–2025